

Bill of Materials — Internal Draft

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User Request

@quotepilot-bom

We need a 24-port access switch for a small retail branch — price sensitivity is high, so budget tier is the priority. I've heard D-Link Enterprise has a competitive SKU in this range. If D-Link doesn't work out for any reason, I'd still like to see the next best budget-tier alternative so the client has something to move forward with.

DATA FRESHNESS WARNINGS

- VEN-004 (Arista Networks): effective_date 2026-02-24 — 76 days old, exceeds 60-day threshold (rule-002). Budget tier SKU V004-SW-0003 affected. Re-confirm pricing and availability before customer delivery.
- VEN-027 (TD Synnex): effective_date 2026-03-01 — 71 days old, exceeds 60-day threshold (rule-002). Premium tier SKU V027-SW-0001 affected. Re-confirm pricing and availability before customer delivery.

RECOMMENDATION

For a price-sensitive retail branch deployment, the balanced tier (Black Box Corporation V041-SW-0001 at \$1,606.72) represents the most defensible single-unit recommendation: it carries no rule violations, ships in 2 days, and holds a 21.0% margin — all without requiring vendor data re-confirmation. If the client's budget ceiling is firm, the budget tier (Arista Networks V004-SW-0003 at \$1,039.90) achieves a 35% cost reduction with a comparable feature profile, but the SE should re-validate Arista pricing before presenting (data is 76 days old). The premium tier (TD Synnex V027-SW-0001 at \$2,285.32) is best reserved for clients where a named-distributor relationship or rapid escalation path adds tangible value. Blended cost across all three tiers is approximately \$1,644 per unit at an average margin of 22.3%; no gap markers were triggered and no abstentions were required for this BOM.

Tiered Recommendations

BUDGET TIER							
SKU	Vendor	Category	Availability	Lead Time	Unit Cost	Margin	Effective Date
V004-SW-0003	Arista Networks VEN-004	networking-switch	In Stock	14 days	\$1039.90	23.3%	2026-02-24
<i>Evidence:</i> catalog: VEN-004/V004-SW-0003, effective_date 2026-02-24 · ruleset: rule-002 (vendor data 76 days old, threshold 60 days — stale-data warning) · ruleset: rule-004 (margin 23.3% — passes 15.0% floor) · ruleset: rule-003 (availability: true — no gate triggered) · ruleset: rule-001 (lead_time_days 14 — within 21-day threshold) · D-Link Enterprise note: V023-SW-0001 excluded via rule-005 (bad data: lead_time_days field has wrong type string '21 days'); V023-SW-0002 and V023-SW-0003 are 48-port 10GbE aggregation switches — do not match 24-port access requirement. No qualifying D-Link 24-port SKU available in catalog. This SKU is the next-best budget-tier alternative per SE instruction.							

BALANCED TIER							
SKU	Vendor	Category	Availability	Lead Time	Unit Cost	Margin	Effective Date
V041-SW-0001	Black Box Corporation VEN-041	networking-switch	In Stock	2 days	\$1606.72	21.0%	2026-03-31
<i>Evidence:</i> catalog: VEN-041/V041-SW-0001, effective_date 2026-03-31 · ruleset: rule-001 (lead_time_days 2 — within 21-day threshold) · ruleset: rule-002 (effective_date 2026-03-31, 41 days old — within 60-day freshness threshold, no warning) · ruleset: rule-003 (availability: true — no gate triggered) · ruleset: rule-004 (margin 21.0% — passes 15.0% floor)							

PREMIUM TIER							
SKU	Vendor	Category	Availability	Lead Time	Unit Cost	Margin	Effective Date
V027-SW-0001	TD Synnex VEN-027	networking-switch	In Stock	2 days	\$2285.32	22.6%	2026-03-01
<i>Evidence:</i> catalog: VEN-027/V027-SW-0001, effective_date 2026-03-01 · ruleset: rule-002 (vendor data 71 days old, threshold 60 days — stale-data warning) · ruleset: rule-001 (lead_time_days 2 — within 21-day threshold) · ruleset: rule-003 (availability: true — no gate triggered) · ruleset: rule-004 (margin 22.6% — passes 15.0% floor)							

Justification

Budget Tier

D-Link Enterprise's only 24-port SKU (V023-SW-0001) was hard-excluded by rule-005 due to a bad-data condition on the lead_time_days field (string '21 days' instead of an integer); D-Link's two remaining catalog SKUs are 48-port 10GbE aggregation units that do not match the 24-port access requirement. Arista Networks V004-SW-0003 is

selected as the next-best budget-tier alternative at \$1,039.90 — the lowest qualifying cost across all 24-port candidates meeting the margin floor.

- SKU V004-SW-0003 (Arista Networks), unit cost \$1,039.90 — lowest qualifying 24-port candidate in the catalog.
- Lead time 14 days — compliant with rule-001 (threshold: 21 days).
- Margin 23.3% — passes rule-004 floor (15.0%); strong margin even at budget price point.
- Rule-002 stale-data warning: Arista Networks effective_date is 2026-02-24, 76 days prior to today (threshold: 60 days) — pricing and availability should be re-confirmed with the vendor before finalizing.
- Vs. balanced tier: \$566.82 cheaper per unit; trade-off is an older data timestamp and a less established reseller channel for this use case.

Balanced Tier

Black Box Corporation V041-SW-0001 sits at the midpoint of the 24-port candidate cost range (\$1,606.72) and carries no rule violations — fully in stock, within lead-time threshold, above margin floor, and data fresh at 41 days old.

- SKU V041-SW-0001 (Black Box Corporation), unit cost \$1,606.72 — midpoint of qualifying 24-port candidate range.
- Lead time 2 days — well within rule-001 threshold (21 days); fastest ship time in this tier.
- Margin 21.0% — passes rule-004 floor (15.0%) with solid headroom.
- No rules fired: data is 41 days old (within 60-day freshness window), fully available, lead time compliant.
- Vs. budget tier: \$566.82 more per unit; gain is a clean catalog record with no stale-data re-confirmation needed.

Premium Tier

TD Synnex V027-SW-0001 is the highest-cost qualifying 24-port candidate at \$2,285.32 and delivers the same 24-port PoE+ managed form factor with a 2-day lead time and 22.6% margin, making it the strongest option for a client who may later need expedited delivery or a higher-tier channel relationship.

- SKU V027-SW-0001 (TD Synnex), unit cost \$2,285.32 — highest qualifying cost among 24-port candidates, representing premium positioning.
- Lead time 2 days — tied for fastest availability in the set, compliant with rule-001 (threshold: 21 days).
- Margin 22.6% — passes rule-004 floor (15.0%).
- Rule-002 stale-data warning: TD Synnex effective_date is 2026-03-01, 71 days prior to today (threshold: 60 days) — pricing should be re-confirmed before customer delivery.
- Vs. balanced tier: \$678.60 more per unit; the incremental cost buys a major distributor relationship (TD Synnex) that may carry additional support, licensing, and services bundling.

Status: **Requires SE review before customer delivery**